

8. (a) (i) The table below shows results of tests carried out on three white Group 1 compounds, **A**, **B** and **C**.

The flame tests were carried out on solids and the silver nitrate tests on solutions of the compounds.

Test	Compound A	Compound B	Compound C
Flame test	red	yellow	lilac
Add silver nitrate solution	white precipitate	cream precipitate	yellow precipitate

Use the information in the table to identify the compounds.

[2]

Compound **A**

Compound **B**

Compound **C**

- (ii) The symbol equation represents the reaction occurring between solutions of silver nitrate and magnesium chloride.

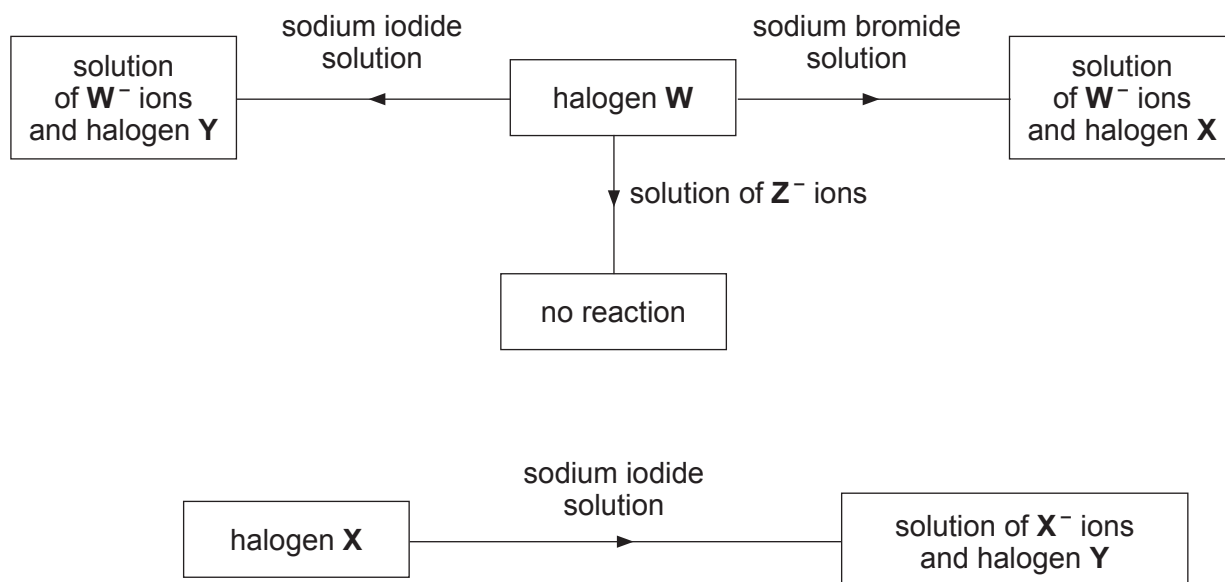


Write the **ionic** equation for the reaction. Include state symbols in your answer. [3]

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- (b) **W**, **X**, **Y** and **Z** represent the halogens fluorine, chlorine, bromine and iodine, but not necessarily in that order.

The diagrams below show some reactions of halogens **W** and **X**.



Use the information to identify halogens **W**, **X**, **Y** and **Z**.

[2]

W

X

Y

Z

- (c) Iron wool burns in fluorine to give iron(III) fluoride.

Write a balanced **symbol** equation for this reaction.

[2]

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8. Sodium is extracted from sodium chloride.

(a) The overall reaction taking place is shown in the equation below.



- (i) When carrying out the reaction 120 kg of sodium chloride was found to produce 38.05 kg of sodium.

Calculate the maximum possible mass of sodium that could be produced and use this figure to calculate the percentage yield of this reaction. [4]

$$A_r(\text{Na}) = 23 \quad A_r(\text{Cl}) = 35.5$$

Maximum possible mass = kg

Percentage yield = %

- (ii) Suggest a reason why the yield is less than 100%. [1]

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- (iii) Suggest why this reaction must be carried out under dry conditions. [1]

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(b) A sample of lithium is found to contain two isotopes.

Isotope	Percentage present in sample (%)
lithium-6	7.59
lithium-7	92.41

- (i) Calculate the relative atomic mass (A_r) of lithium. Give your answer to **three** significant figures. [3]

$$A_r = \frac{(\text{isotope 1 mass} \times \text{abundance}) + (\text{isotope 2 mass} \times \text{abundance})}{100}$$

$A_r =$

- (ii) State the difference between the atomic structures of lithium-6 and lithium-7. [1]

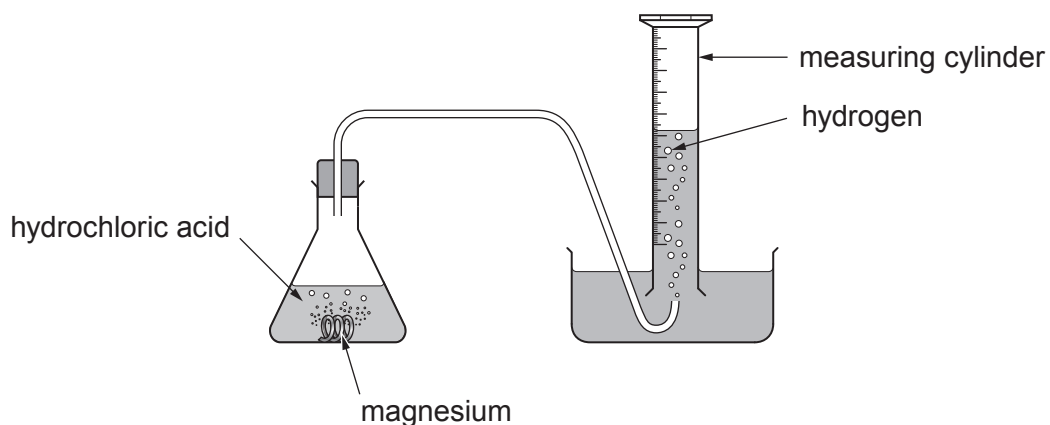
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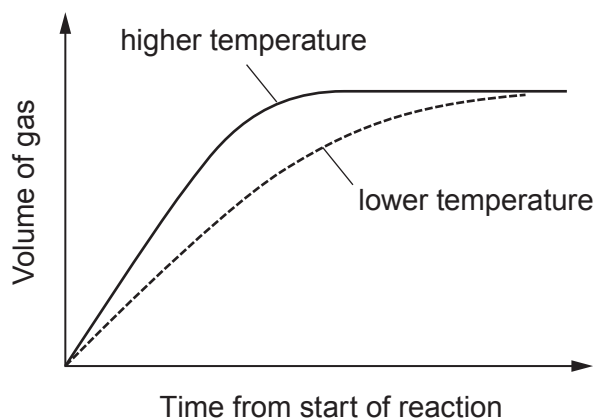
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8. The following apparatus can be used to investigate the rate of the reaction between magnesium and excess dilute hydrochloric acid.



The results obtained at two different temperatures are shown below.



- (a) Explain the results obtained at different temperatures in terms of particle theory. [3]

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(b) Explain the change in rate over time. [2]

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(c) State **one** way of improving the validity of the results obtained. Explain your answer. [2]

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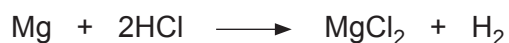
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(d) The equation for the reaction taking place is as follows.



(i) In the experiment, 0.445 g of magnesium was used.

Calculate the mass of hydrogen that will be produced during the reaction. [3]

$$A_r(\text{Mg}) = 24$$

$$A_r(\text{H}) = 1$$

Mass = g

(ii) It is known that one mole of gas has a volume of 24 dm^3 .

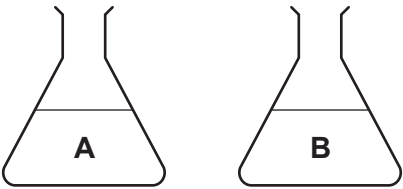
Use the equation below to calculate the volume of hydrogen produced. [2]

$$\text{number of moles} = \frac{\text{volume of gas}}{24}$$

Volume = dm^3



8. Hard water can be softened using different methods, including boiling the water and passing it through an ion exchange column.
- (a) Two hard water samples, **A** and **B**, were tested for hardness, using soap solution. Both samples were tested before and after boiling, and after passing through an ion exchange column.



	Volume of soap solution needed to produce a lather (cm ³)		
Water sample	Before boiling	After boiling	After ion exchange
A	15	15	2
B	17	9	2

State the type (or types) of hard water that each sample contains. Give your reasoning. [4]

A

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B

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- (b) Sodium carbonate (washing soda) can also be used to soften hard water. It reacts with the magnesium compounds in the water.

Write a balanced symbol equation for the reaction between sodium carbonate and magnesium chloride, to form sodium chloride and magnesium carbonate. [3]

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8. (a) The table shows information about some Group 7 elements.

Element	Radius of the atom (nm)	Number of electrons in outer shell
chlorine	0.099	7
bromine	0.114	7
iodine	0.133	7

- (i) State how the radius of the atom changes down Group 7. [1]

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- (ii) Explain in terms of electronic structure why the reactivity of the elements decreases down Group 7. [2]

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- (b) Sea water contains sodium bromide, which is the raw material for the production of bromine.

Between 1953 and 2004 bromine, Br₂, was manufactured from sea water at a chemical plant in Amlwch, Anglesey.

Sea water was treated with chlorine, which converts sodium bromide into bromine.

Complete the balanced equation for the reaction between chlorine and sodium bromide. [3]



(c) 4.47 g of copper bromide was found to contain 1.27 g of copper.

(i) Calculate the mass of bromine in 4.47 g of the copper bromide. [1]

Mass = g

(ii) Calculate the simplest formula of the copper bromide.

You **must** show your working. [2]

$A_r(\text{Br}) = 80$ $A_r(\text{Cu}) = 63.5$

Simplest formula

